Flood Risk Assessment

Application reference: 91.48267.FUL
Roseacre House, Barnby Dun, Doncaster, ZZ38 5BX

Terms of Reference

This report was commissioned by the applicant to support a planning application for a non-residential storage building/barn at Roseacre House, Barnby Dun, Doncaster, ZZ38 5BX. The site is located to the north of Barnby Dun which is to the north of Doncaster. The development site lies within Zone 3 of the Environment Agency Flood Map (version 2.8.2), being the zone with risk of 1 in 100 year (1% Annual Exceedence Probability) or greater for river flooding and 1 in 200 year (0.5% AEP) or greater for tidal/coastal flooding. The development site is within an existing developed area and is less than 1 hectare. It is usual for the Agency to raise an objection to development applications within the floodplain or Zone 2 or 3 of the flood map until the question of flood risk has been properly evaluated. The Agency will also object to developments where the total site area is in excess of 1 hectare until suitable consideration has been given to surface water runoff.

Approach to the Assessment

As there are three sources of flood risk - The River Don, local watercourses and surface water runoff - it is necessary to determine flood water levels at the site for the desired return periods emanating from these sources. Consideration has also been given to the site flooding from either overland flow or ponding of localised rainfall within the site. A strategic flood risk assessment was undertaken on behalf of Doncaster MBC and has been used to inform this assessment. The River Don is located approximately 530m to the north west of the proposed development site. The Environment Agency does have modelled flood data which may assist in predicting the design flood level for the river adjacent to the proposed development site. There are several local watercourses to the south and east of the development site. The Environment Agency did not provide any modelled flood data for the watercourses. However, the surrounding local watercourses discharge into the River Don downstream of the site. Subsequently, the flood levels within the River Don will influence local flood levels within the local watercourses. It is considered that the River Don will produce the highest flood levels and greatest impact on the development site. As such, no further consideration has been given to this mechanism. The proposed development is for a replacement non-residential storage building/barn. The existing building is supported by roof drainage. There are no reports of sewer incapacity local to the site. The client provided a topographical survey for the site. The existing ground floor level of the building is approximately 5.30mOD. The requirements for flood risk assessments are generally as set out in National Planning Policy Framework (NPPF). The detail and complexity of the study required should be appropriate to the scale and potential impact of the development. For the purposes of this study, the following have been considered:
- Available information on historical flooding in the area.
- Site level information.
- Details of structures, which may influence hydraulics of the watercourse and consideration of the effect of blockage of structures.
- Estimates of design levels, equivalent to a 200-year (coastal/tidal) and a 100-year (fluvial) return period flood event.
- Allowances for increased flows resulting from the effects of climate change.
- Allowances for sea level rise resulting from the effects of climate change. Assess the existing runoff characteristics and the potential impact the proposed development will have on the runoff. Further guidance is also provided in the CIRIA Research Project 624 “Development and Flood Risk: Guidance for the Construction Industry”.

Application of Sequential & Exceptions Test

The development site lies partly within Zone 3 of the Environment Agency Flood Map (version 2.8.2), being the zone with risk of 1 in 100 year (1% AEP) or greater for river flooding and 1 in 200 year (0.5% AEP) or greater for tidal/coastal flooding. The proposed development is non-residential storage building/barn and as such considered to be less vulnerable. The proposed development is for the redevelopment of existing buildings. In accordance with the Doncaster Metropolitan borough Council Planning Guidance the proposed development is exempt from the sequential test. The guidance states: Proposals for demolishing and replacing existing buildings (other than housing market renewal areas) will not require a Sequential Test.

Site details

Application reference: 91.48267.FUL.
Site: Roseacre House, Barnby Dun, Doncaster, ZZ38 5BX.
Existing land use: Residential.
Proposed development: Non-residential storage building/barn.
County: South Yorkshire.
Local planning authority: Doncaster Metropolitan Borough Council.
Internal drainage board: Not applicable.

Site Description

The proposed development site currently consists of a building and semi permeable access track and vegetated areas. The total site covers an area of approximately 2000 m². The land within the site boundary lies at a level of between 4.9mOD and 5.2mOD. Roseacre House, Barnby Dun, Doncaster, ZZ38 5BX adjacent to the site entrance is at a level of 5.25mOD. The existing ground floor level of the dwelling is 5.30mOD. There is existing residential development to the north east of the site. There is a positive drainage system from the buildings which is believed to convey flows into the local drainage system. The proposal is to provide a non-residential storage building/barn. It is considered that the existing footprint will be increased from 100m² up to approximately 170m².

Past Flooding History

A search of the British Hydrological Society Chronology of British Hydrological Events identified records of past flooding within the Barnby Dun area of Doncaster. An internet search for flooding in the area provided no further information.

Reference: British Hydrological Society, Chronology of British Hydrological Events, University of Dundee.

SFRA Flooding History

The Strategic Flood Risk Assessment contains references to previous flooding in the area.

Environment Agency Flooding History

The Environment Agency supplied historical flood maps recording previous flood events. These record flooding in the area local to the site.

Reservoir Flood Risk

The site is located within an area which is at risk from reservoir breach failure. The River Don will act as a conduit for the reservoirs within the peak district. The likelihood of a flood is extremely low.

Surface Water Flood Risk

The site is located within an area which is at a low risk from surface water flooding. It is considered that surface water flooding in the area of the site would be associated with the River Don flooding.

Possible Flooding Mechanisms

As there are two sources of flood risk - The River Don and onsite runoff - it is necessary to determine flood water levels at the site for the desired return periods emanating from these sources. The River Don which is the responsibility of the Environment Agency and is located approximately 530m to the north of the proposed development. The area is defended by flood embankments, therefore overtopping and breach will require consideration. The proposed development will increase the impermeable area within the site. Consideration will need to be given to the existing drainage route and the drainage characteristics in order to evaluate the impact that surface water runoff from the site will have on the site and elsewhere. Due to the local topography the site itself is relatively flat with a gentle gradient in several directions. There are no depressed areas within the site which could encourage ponding. There are no areas of higher ground close to the development which could encourage overland flow towards the site. As such both of these mechanisms have not been considered further. Yorkshire Water is the statutory water undertaker and is responsible for the public sewer systems within the Barnby Dun area. Yorkshire Water maintains a register of historical sewer flooding events (DG5 Register) within the area. There are no reported incidents close to the site. The SFRA provided no further information.

Requirements of the Environment Agency

The Environment Agency, as part of its development control procedures, generally require finished floor levels to be set above the 1% AEP plus climate change flood water level at the site. The development is residential in nature, as such it is considered that access and egress from the development site will be essential during times of extreme floods.

River Don

Approximately 530m to the north of the development site is the main channel of the River Don. The River Don is considered to be fluvial at this point and as such the required standard of defence is approximately 1 in 100 years as the crest level is surveyed at between 7.33mOD and 7.67mOD by the Environment Agency. The Environment Agency commissioned a hydraulic model of the River Don system following previous floods. The model was constructed using ISIS software and was subsequently updated following further flooding. Further modelling of the River Don was undertaken, and the most recent available model data have been used in this assessment.

Overtopping of the defences during 1 in 100 year event

The 1 in 100 year level provided at the River Don model point is 7.7mOD based on defended scenario. The undefended scenario estimates a flood level of 5.26mOD. The right bank level of the flood embankment is recorded at a minimum of 7.33mOD which is lower than the 1 in 100 year flood level. The existing ground floor level of the buildings within the existing site lies at a level of 5.3mOD with adjacent ground levels of approximately 5.2mOD or less. The Environment Agency also provide flood mapping of the event which shows that the site remains dry. Consideration will need to be given to the effects of climate change over the design life of the development, a breach of the defences and overtopping during an extreme flood event (1 in 1,000 year).

Increase in estimated flood level due to Climate Change

As the proposed development includes residential land use only, consideration has therefore been given to take into account the potential effects of climate change over the next 100 years in accordance with NPPF. Current guidance recommends that an increase of 30% are applied to non residential developments within the Humber Catchment. The 1 in 100 year plus climate change 30% level provided at the River Don model point is 7.77mOD based on defended scenario. No undefended scenario estimates were provided. The right bank level of the flood embankment is recorded at a minimum of 7.33mOD which is lower than the 1 in 100 year plus climate change 30% flood level. The 1 in 100 year plus climate change 50% level provided at the River Don model point is 7.78mOD based on defended scenario. No undefended scenario estimates were provided. The right bank level of the flood embankment is recorded at a minimum of 7.33mOD which is lower than the 1 in 100 year plus climate change 50% flood level. The existing ground floor level of the buildings within the existing site lies at a level of 5.3mOD with adjacent ground levels of approximately 5.2mOD or less. The Environment Agency also provide flood mapping of the event which shows that the site remains dry.

Overtopping of the defences during 1 in 1,000 year extreme event

Consideration has been given to the extreme flood event and its likely impact on the site and its occupants. The 1 in 1,000 year level provided at the River Don model point is 7.79mOD based on defended scenario. The undefended scenario estimates a flood level of 5.64mOD. The right bank level of the flood embankment is recorded at a minimum of 7.33mOD which is lower than the 1 in 1,000 year flood level. The existing ground floor level of the buildings within the existing site lies at a level of 5.3mOD with adjacent ground levels of approximately 5.2mOD or less. The Environment Agency also provide flood mapping of the event which shows that the site remains dry.

Breach Failure of the defences

Consideration has been given to a breach failure during a 1 in 100 event which is similar to the existing crest level of flood defence. The following flood levels in line with the site have been used within the breach modelling exercise. - flood level = 7.36mOD - Crest height of existing flood defence in line with site = 7.36mOD The current flood defences approximately provide the 1 in 100 year standard of defence. This scenario would require the river level to be just below the top of the flood embankment directly in line with the site. The likelihood of this flood event occurring is considered to be less than 1%. This flooding mechanism requires a failure of the flood embankment to also occur simultaneously with the high water level. As the flood embankment are regularly inspected and well maintained the likelihood of this is considered to be low. The landward side of the defences is 500m from the existing building within the development site. Therefore, the risk of breach failure has been initially assessed by adopting the simple approach as recommended in DEFRA report FD2320/TR2 and utilising table 12.2 of that report, the danger classification is considered to be “danger for most” as the building is 530m from the river and the flood water depth at the site is approximately 2.06m (7.36mOD - 5.3mOD existing site level at building). Danger for most classification includes the young, elderly and infirm. As such, a more detailed assessment of the breach will be required in line with the site. The more detailed assessment will assume that there are no buildings or infrastructure between the river defences and the site which could absorb or deflect breach flood flows away from the site. Hence, a worst-case scenario will be analysed.

Breach equivalent to the River Don 1 in 100 year level

An indication of the flood risk area can be made by a simple breach analysis. Assuming that flow through a breach can be represented by the broad-crested weir equation: Q = Cd B h1.5 Where: Q = flow through the breach (m3/s) B = width of the breach (m) h = depth of water through the breach (m) Cd = coefficient (usually between 0.5 and 1.7) The 1 in 100 year flood level (7.36mOD) adjacent to the proposed development is 2.36m above existing ground level to the rear of the flood embankment (5mOD), hence the maximum breach height would be 2.36m during the 1 in 100 year flood event and h has therefore been taken as 2.36m, Cd has been assumed to be a conservative 1.5. Putting these values into the weir equation indicates the following flows for a breach of 50m wide, which is considered to be appropriate width for embanked flood defences by the Environment Agency. - 1 in 100 year, h = 2.36m flow through breach = 272m³/s

Rapid Inundation Zone Calculation

The depth of flooding such flows would generate are dependent on many factors, but a simple assumption based on trigonometry is that the maximum width of flow (w) a distance (a) from the breach is reached when w = Q. This gives the critical distance from the breach where flow depth would be significant as: a = (Q - b) / 2 Using this equation, the critical distances for flooding from a breach of 50m is 111m during 1 in 100 year scenario. The building is approximately 530m from the River Don defences. Given the conservative nature of these calculations, this indicates that the building is outside the point where the flood depth and the flood velocity would be significant; hence the site is potentially not within the rapid inundation zone.

Breach Model

The proposed building is located 530m from the landward side of the flood embankment. As the flood water moves inland from the breach it will disperse in all direction reducing the flood depth. As the water travels across the ground the friction losses will reduce the velocity. The model assumes that the flood water will disperse at a 45 degree angle from the breach; therefore, once the flood water reaches the end of the site which is 500m from the flood embankment the width of the flood flow will be 1050m. In order to evaluate the impact of the breach on the site an assessment of the water depth and velocity at the site need to be made. This has been done by simply constructing four cross sections with a HEC RAS version 5.0 hydraulic model. The first cross section represents the breach in the flood embankment and the second is the bottom of the flood bank. The third section is located within open fields. The fourth and fifth sections are either side of the proposed building. The seventh section is within open fields to the north of the site. The ground is a mixture of agriculture and grass therefore a Manning's value of 0.035 has been conservatively assumed. The model estimates that during the breach scenarios the following flood depths and velocities would occur within the site. For the 100-year scenario with river level 7.36mOD, the model results are: CS450 - ground level 5.30mOD, water level 5.85mOD, depth 0.55m, velocity 0.43m/s; CS500 - ground level 5.30mOD, water level 5.88mOD, depth 0.58m, velocity 0.45m/s. The largest flood depth occurs at cross section 500 with a depth of 0.58m and a velocity of 0.45m/s during a 1 in 100 year event. Therefore, the area proposed for the buildings currently located within an area which is danger to most. It should be appreciated that the peak flood depth of 0.58m and velocity of 0.45m/s will be achieved immediately following a breach and may only be sustained for several minutes as the flood water on the landside of River Don ponds thus reducing the flood velocity and depth.

Surface-water runoff - existing drainage

The existing site has an area of approximately 0.2 hectares and is dense vegetation. Most of the surface is unpaved and simply infiltrates the subsoils.

Proposed Impermeable Area

The proposed development is a replacement single non residential building and associated driveway which is unlikely to exceed 0.02 hectares in roofed and paved area. Therefore, the proposed development is likely to have a minor impact on the local drainage system. The following options should be considered for the proposed development.

Soakaways

Soakaways are a type of infiltration device, and are considered to be a simple way of dispersing surface water from developments where connection to an existing surface water drainage system or watercourse is impractical or unwarranted. The basic principle is that of a 'reverse well' and is greatly dependent upon ground conditions within the development site. Soakaways allow water to infiltrate back into the ground, recharging ground water, reducing and treating surface water run-off. Excavations previously undertaken within the site have revealed free draining sand to over 2m with no groundwater presence. It is recommended that a site investigation is carried out to ascertain the nature of the ground conditions if this form of drainage is to be utilised.

SUD's (Sustainable Urban Drainage)

Sustainable Drainage Solutions (SUDS) are an alternative to the traditional approach of collecting stormwater in pipes and discharging it into sewers or watercourses. The Environment Agency will require that surface water runoff is controlled as near to its source as possible, through a sustainable approach to surface water management (SUDs). This approach utilises a range of techniques such as rainwater harvesting, infiltration trenches, detention ponds and swales to reduce flood risk by attenuating the rate and quantity of surface water runoff from a development site. Similar to soakaways, infiltration trenches are greatly dependent upon the ground conditions within the site.

Adjacent open watercourse

Piped discharge to the adjacent local watercourse at the existing greenfield rate. As such, attenuation will be required in the form of shallow ponds, oversized pipes or below ground storage such as crates.

Raising Floor Levels/Land Raising

The River Don is inline with the site and is considered to represent the greatest flood risk to the site. The River Don estimated 1 in 100 year flood level is 7.7mOD. The estimated 1 in 100 year plus climate change 50% flood level is 7.78mOD. The left bank of the River Don flood defences is at a level of at least 7.33mOD. As such the 1 in 100 year plus climate change flood level will not over top the flood defences in line with the site. The Environment Agency also provide flood mapping of these events which shows that the site remains dry. However, as the site is 530m from the River Don it is considered that the site is located within an area which is danger to most if the defences were to breach. It is estimated that during a breach of the defences that the proposed building (5.3mOD) would be flooded to a depth of 0.58m (5.88mOD) and a velocity 0.45m/s. The proposal is for a replacement non residential building; therefore, habitable ground floor levels should be raised to at least 5.9mOD. It is further recommended that flood proofing and flood resilience measures are used to provide an additional 500 mm of mitigation.

Emergency Access & Egress

It is considered that during a 1 in 100 year and the extreme 1 in 1,000 year events which result in the defences being overtopped that the proposed development is within an area which is considered to be low hazard and therefore accessible to all users. During a 1 in 100 year breach event it is considered that the site is located in an area which is danger to most and the flood depths could be over 0.5m within the access road. The ground floor level of the building will be a safe area above the flood water. The ground and first floor level will incorporate a bathroom and as such facilities will be available during the flood period. It is also recommended that electrical junction boxes are located above the estimated flood level plus 600mm to ensure that the power is available for flood events which last several days. It is considered that a breach could remain open for up to 24 hours whilst the Environment Agency undertake the necessary earthworks to close the breach before another high tide flows through the breach. It is considered that the water level may remain at a high level for another 24 hours once the breach is closed as the local drainage system evacuates flows back into the River Don. As such it is highly probable that anyone using the first floor safe area during a breach event could be within the building for at least 48 hours. Although the site is located within an area which is “danger to Most”, the South Yorkshire Fire & Rescue services considered that the current Met Office & Environment Agency flood warning arrangements should provide the appropriate notification of a potential flood situation.

Flood Resilience Measures

Consideration should be given to flood proofing the building to a level equivalent to the estimated breach level to reduce the residual damages if an extreme flood was to occur. Flood proofing is a technique by which buildings are designed to withstand the effects of flooding. There are two main categories of flood proofing, which are dry proofing and wet proofing. Dry proofing methods are designed to keep water out of the building, and wet proofing methods are designed to improve the ability of the property to withstand the effects of flooding once the water has entered the building. Both would be required in this case, due to the possible failure of the dry proofing methods. Where wet proofing is required it is important that a flood response plan should be prepared and practised regularly, so that any contents of the building can be moved to design flood level if required or are built to withstand immersion in water or are designed to be easily replaceable. The differential pressures across load bearing walls and the flotation effect that will occur during flood events should be taken into account when considering dry proofing techniques. For most existing properties this means that dry flood proofing should only be considered if the expected flood depth is under 0.6m, which in this particular case is expected to be less than 0.5m in depth and as such dry proofing is likely to be successful. It is therefore considered that flooding in excess of 0.6m will cause some damage to the building; however, this is considered that the proposed redevelopment will provide considerable betterment above the existing building. The following flood-proofing measures can be incorporated within the building design.

External Walls

Careful consideration of materials: use low permeability materials to limit water penetration if dry proofing required. Avoid using timber frame and cavity walls filled with water absorbing materials. Consider applying a water resistant coating. Provide fitting for flood boards or other temporary barriers across openings in the walls.

Internal Walls

Avoid use of gypsum plaster and plasterboards; use more flood resistant linings (e.g. hydraulic lime, ceramic tiles). Avoid use of stud partition walls.

Floors

Avoid use of chipboard floors. Use concrete floors with integrated and continuous damp proof membrane and damp proof course. Solid concrete floors are preferable; if a suspended floor is to be used, provide facility for drainage of sub-floor void. Use solid insulation materials.

Fitting, Fixtures and Services

If possible, locate all fittings, fixtures and services above design floor level. Avoid chipboard and MDF. Consider use of removable plastic fittings. Use solid doors treated with waterproof coatings. Use solid wood staircases. Avoid fitted carpets. Locate electrical, gas and telephone equipment and systems above flood level. Fit anti-flooding devices to drainage systems.

Demountable flood defences

External doorways should be fitted with demountable flood barriers which are 500 mm high. This will reduce the ingress of flood water significantly. Reference: Development and Flood Risk Guidance for the Construction Industry, CIRIA C624, London.

Flood Protection Measures

Keeping water out of the buildings, or limiting the ingress of floodwater, is recommended when considering flood protection measures. Excluding water will help to reduce damage to the internal fabric of the building and its contents. Such measures are referred to as dry-proofing and include: - Temporary flood barriers. - Measures to reduce seepage through walls and floors. - The installation of non-return valves on sewers. Movable flood barriers can be very effective in preventing or reducing the volume of floodwater entering through doors and other external openings in walls, such as windows and airbricks, as long as a good quality product is installed in accordance with the manufacturer's instructions. Although barriers may not totally prevent the ingress of water into a building, as seepage may still occur through the floor or walls, they can provide valuable time in which to move people, vehicles, expensive equipment and other essential items to higher levels before floodwater rises inside the building. Flood barriers on wall openings can also reduce the amount of contaminated silt and debris entering the property. Water that seeps through the ground or walls is likely to be filtered to some extent and therefore is usually cleaner than floodwater entering larger openings such as gaps around doors and airbricks. Although temporary barriers have a number of benefits they rely on the occupants taking positive action in good time before the floodwaters arrive. If the barriers are not erected quickly enough following a flood warning, such temporary measures will provide no protection against flooding. There are many suppliers and various types of flood doors and barriers which may be utilised at the development site. An alternative and cost effective method of flood protection to the ground floor entrances involves the use of flood boards and sandbags. In order for this method of flood protection to be effective, the pointing on the lower part of the buildings must be sound and air bricks situated below possible flood level should be temporarily plugged up. A piece of 18 mm marine plywood cut to fit across the door jambs may be utilised as a temporary flood board. The plywood can be nailed in place if required and sealed with clay, plasticine, rubber, carpet or underlay, or wadded with wet newspaper, to facilitate holding back the flood water. In addition, heavy duty polythene sheeting may be stapled over the doorway on the outside of the building, or fixed with drawing pins or tacks to the jambs and door sill. The door must be closed to accommodate the weight of flood water, but this method will help seal the gaps under and around the door opening. Filled sandbags may be used to prevent flood water from flowing through air vents, doors and other water accessible points into the building. There are two sandbag types which include hessian sacking, or polypropylene bags. Hessian sacks should be stored empty, in a dry place and should be filled just before they are required; otherwise deterioration of the sacking material may occur over a short period of time. When properly stacked sandbags are the most efficient method at stopping water ingress, however they tend to decay once filled with sand. Polypropylene sandbags are more durable when filled but are considered to be less efficient in their application than traditional sandbags.

Flood Warning

It is highly recommended that the residents of the proposed dwelling are advised to sign up to Floodline Warnings Direct. The service is able to provide flood warnings via telephone, mobile, fax or pager. More information on the likely timing and extent of warnings can be obtained by request from the Environment Agency, via their website or obtained by their 'Quickdial' recorded information service.

Evacuation Plan

For health and safety reasons there is a preference to evacuate, as failure to do so, would put residents and others (e.g. rescuers) at risk of injury. However, should evacuation not be possible the first floor level within the main building will ensure that the building is above the estimated breach flood level at the site. It should also be noted that all sleeping accommodation in the main building is located at first floor level. The converted garage will have a single bedroom at ground floor level which will be elevated above the breach level. The care home operator should be made aware of the potential for overtopping of the flood defences and the likely effect this could have on the development especially during events which exceed the extreme flood event (+1,000 year event). It is highly recommended that the care home operator is advised to formulate an evacuation route for the site. The Flood Warnings Direct Service provides flood warnings directly by telephone, mobile, fax or pager and also provides practical advice on preparing for a flood, and what to do if one happens. During evacuation, residents should relocate south towards Flood Zone 1 within Barnby Dun.

Conclusion

It is concluded that there is a risk of flooding due to overtopping and breach from the River Don. However, it is considered that the redevelopment will provide betterment by the development of a flood warning and evacuation plan and the inclusion of flood proofing and flood protection.